Engine Model(s): PT6A

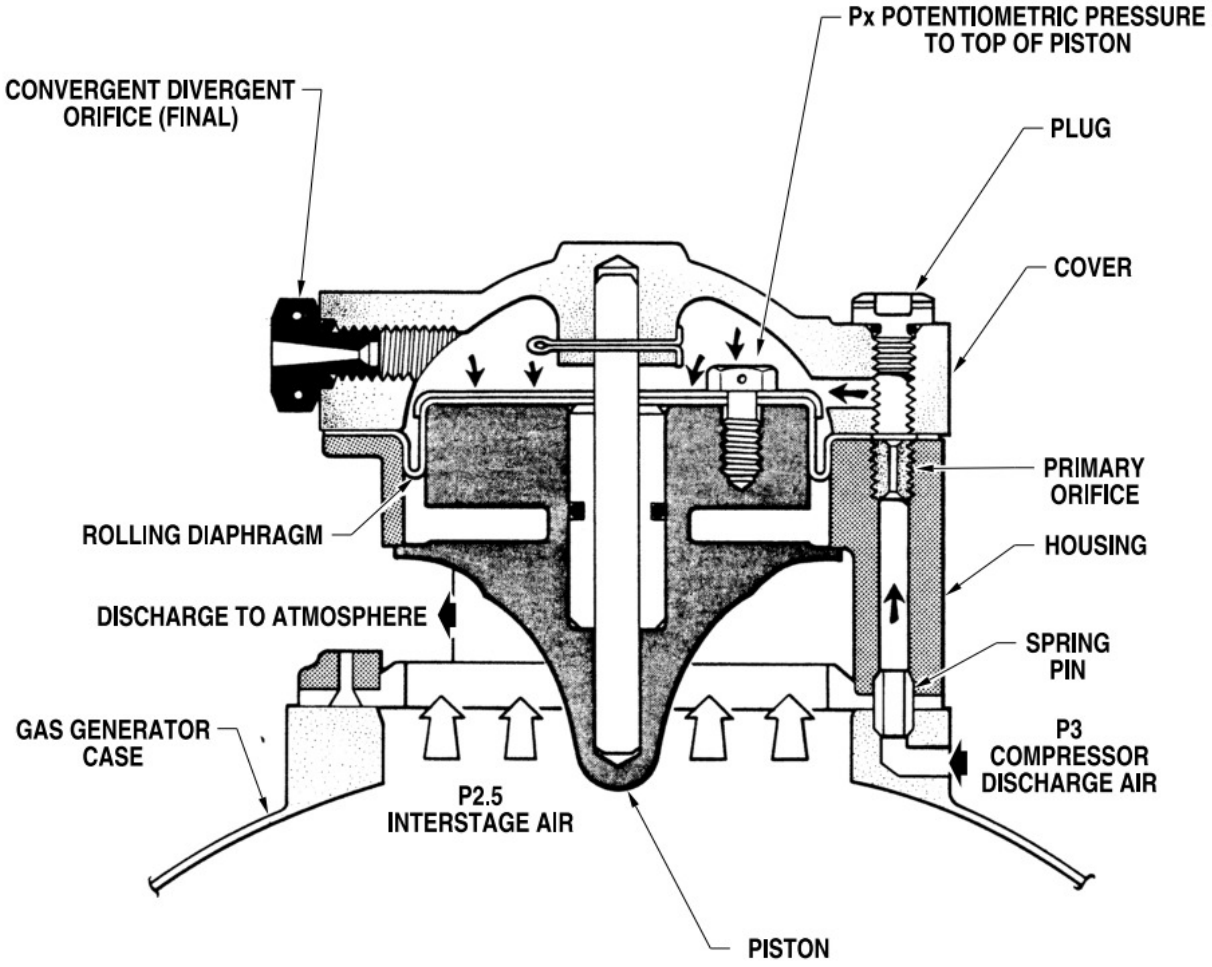
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COMPRESSOR BLEED VALVE - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. Fig. 1)

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Figure 1 Compressor Bleed Valve - Cross-Section



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The compressor bleed valve, located at the 7 o'clock position on the gas generator case, consists of a piston type valve operating in a ported housing. The piston is supported in the bore by a rolling diaphragm and a guide pin which permits full travel of the piston in either direction, to open or close the ports, while at the same time effectively sealing the air chamber at the top of the piston.

The bleed valve is bolted to the ported flange on the gas generator case. A hollow spring pin protruding from the valve mounting flange serves to align the compressor discharge (P3) air hole in the gas generator case flange with the bleed valve P3 passage.

P3 air is taken through a primary metering plug to a convergent-divergent metering orifice, which vents to ambient atmosphere. The convergent-divergent orifice is designed to create a choked flow over the necessary engine operating range, resulting in a fixed pressure ratio P3/Px, where Px is the control pressure between the primary metering plug and theconvergent-divergent orifice acting on the bottom of the piston. The position of the floating piston is dependent on the balance of forces between compressor interstage air P2.5 and Px. At low engine power, where P2.5 is much greater than Px, the valve is fully open. As power is increased, the force from Px pressure increases and balances the force from the P2.5 pressure, and the piston starts to close. The valve does not snap shut since P2.5 pressure increases as the valve is closed. This results in a progressive closing at constant P3/P2.5. At high power the valve is completely closed and Px is higher than P2.5. As this is a pressure ratio control, it will maintain constant P3/P2.5 when it is modulating, regardless of altitude and ambient temperature.

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